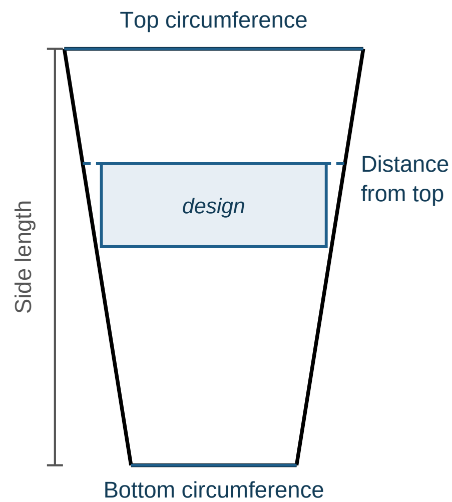
Tapered Cup Etching Pattern — How to Use

Turns a photo or logo into a laser-etching pattern for the front-facing panel of a tapered cup or glass, pre-warped so it looks undistorted — not pinched at the edges — when the finished, curved glass is viewed head-on. Etched with a rotary attachment.

1. Measure your cup

- **Bottom circumference** & **top circumference** — wrap the tape around the rim at the bottom and top of the design area (not necessarily the whole cup).
- **Side length** — lay the tape flat along the tapered side, bottom rim straight up to top rim.
- **Distance from top** — same kind of measurement, from the top rim down to where the design should start (0 if it starts right at the top).



2. Upload & enter your numbers

Upload a PNG or JPG, then enter the four measurements above plus a **design width** (how wide the finished etching should *look*, viewed head-on — the image is scaled to this, keeping its own proportions; height is never entered directly). Switch the **mm / in** toggle at any time — every field converts automatically. Click any **?** icon for that field's own explanation.

3. Check the live preview, then generate

The panel below the inputs updates as you type: design height, the diameter at the design's own position, front coverage, and the pattern's true physical width (see the box below). Pick a resolution (DPI) and whether to dither (recommended for photos; leave off for high-contrast logos/line art), then click **Generate pattern**.

4. Verify, then download

Click **View in 3D** to rotate the actual cup shape and confirm the design reads correctly from the front; toggle the semi-transparent original-image overlay to compare it directly against the curved, warped pattern. When it looks right, click **Download**.

Setting up your rotary job — read this

The tool reports two numbers that are **not** what you typed — both matter for the physical setup:

• **Rotary calibration diameter** — the diameter *at the design's own position*, not necessarily your cup's overall diameter. Enter this in your rotary attachment's own calibration.

• **Pattern's true physical width** — wider than the "looks … wide viewed head-on" design width you typed (peeling any design off a curved surface always yields more material than its straight-line width). Enter *this* number, not your design width, as the image width in your rotary job.

Tips: keep design width comfortably under the reported diameter (roughly under 60–65% of it) for a result that's forgiving of small real-world setup imperfections — designs that push close to the diameter need much more aggressive correction and are far more sensitive to the cup being perfectly level and the viewer being straight-on. Nothing you upload is stored beyond your session.